

Masons Training Experience Sharing Session

9th December 2016
NRA Conference Hall

Presentation Contents

- Summary of information collected by HRRP district teams:
 - Training of Trainers
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TRAINERS PERSPECTIVE

Information Collected

- Where did you complete the ToT? What was the duration of the ToT?
- Since completing the ToT have you delivered any training courses? If yes:
 - How many, what curriculum, and how many masons were participating in the course?
 - Were you lead trainer? Assistant Trainer? Observing?
 - Do you feel that you have had sufficient support to conduct these trainings?
- What have been the major challenges you have faced in delivering training courses?

Trainer Development

- Trainers interviewed = 32
- Trainings these trainers have completed as:
 - Observer: 12
 - Assistant Trainer: 13
 - Lead Trainer: 206
- ToTs these trainers completed as:
 - Observer: 1
 - Assistant Trainer: 5
- Many trainers moved straight from participating as trainee in ToT to being lead trainer – quality issues?

Challenges – 7 Day

- Accessing materials for the practical sessions
- Duration of the training — too short!
- Training too theoretical, materials relied too heavily on text
- Insufficient focus on carpentry
- Selection of masons:
 - New masons also selected to take part; mixed levels of experience
 - Some did not work as masons after the training
- Allowances are difficult and allowances major motivation for some participants. Practicing and interested masons may participate if no allowances provided.
- Insufficient support provided for trainer to adequately deliver training course

Challenges - OJT

- Participants were new to construction and struggled to keep up
- Full participation in 50 days vocational training is challenging, participants missed days, could not attend for various reasons, etc.
- There was confusion around some of the technical aspects promoted by the training

Recommendations

- Demonstration houses / buildings should be constructed during or immediately after training sessions, to support masons and serve as a long term reference.
- 7 days training is not sufficient. Regular guidance, supervision, and follow up in ground are also important to ensure compliant construction.
- Community awareness programmes are also important to ensure adoption of earthquake resilient techniques.
- Provide references or booklets to masons after training so they can review what they learnt.



TRAINING FOR EXISTING MASONS: PARTICIPANTS PERSPECTIVE

Information Collected

- What was the duration of the training course?
- Who conducted the training?
- How long had you been working as a mason before the earthquake?
- Were any of the buildings you worked on damaged? If yes:
 - How did you feel about this?
 - Did the training you completed help you to understand what could have been done differently to prevent that?
 - If yes, what?
- How much do you charge households for your work (daily rate)?
Have you increased your daily rate now that you have been trained?
- What are the challenges that you face in implementing the earthquake resistant construction elements covered in your training course?
- Are there any topics that your training didn't cover that you feel it should have? What are these?

Daily Rates

- **Gorkha:** Ranged from 500-800 NPRs / day, and had increased by 200-300 NPRs
- **Makwanpur:** Ranged from 600 – 900 NPRs / day, with half reporting that their rate had not increased and half reporting an increase of 200-300 NPRs
- **Nuwakot:** Ranged from 900-1500 NPRs / days, and had increased by 200-300 NPRs
- **Okhaldhunga:** 600 NPRs / day with no increase in rate

Daily Rates – Contd.

- **Dolakha:** Ranged from 800-1500 NPRs / day, with all reporting increase due to high demand for skilled labour and high cost of materials. Increase in daily rate ranged from 300-700 NPRs.
- **Kavre:** Ranged from 700-1200 NPRs / day. Some reported no increase, but if it had the increase ranged from 200-300 NPRs.
- **Dhading:** Ranged from 700-1000 NPRs / day, most reported no increase in rate after training. Where there was an increase it was between 200-300 NPRs.
- **Rasuwa:** Ranged from 800-1200 NPRs, with increases of between 200-400 NPRs
- Changing the rate based on whether or not food and lodging is provided is common

Pre-EQ Experience

- Number of years of experience ranged from none (!) to 40 years
- Many reported that damage to buildings built before the earthquake was due to the materials used, especially stone and mud mortar
- Many reported that they had not used bands, through stones, etc. before the earthquake, but now understood the importance of these elements
- Some had been working in other districts which were not affected by the earthquake
- Some indicated that they hadn't received any complaints about the damage to the buildings they had worked on

Challenges

- Additional time to build a house with EQ resistant elements
- Difficult to source good quality stone
- Older masons don't want to change their attitude and continue to apply old techniques
- Trainers not very good; training not based on local context
- Households don't want to spend the money required to include the EQ resistant elements in their house
- Timber is not available
- Ratio of 1:3 (Cement:Sand) is being adopted where previously 1:5 or 1:6 was used, but house owners want to use 1:5 or 1:6 to reduce costs.
- In the villages it is not practically possible to build using the EQ resistant elements

Recommendations

- Training should also cover concrete blocks, RCC, carpentry, and roofing
- More practical sessions, include time on learning to read drawings, design of new buildings, construction work in hilly and geographically difficult places
- Overall duration not sufficient to learn in detail about the EQ resistance techniques
- Refreshing trainings would be helpful
- Before providing certificates the trainees should have to take an exam to check that they have learnt all the key elements of the course

Legend

- International Boundary
- District Boundary
- VDC/Municipality Boundary

Target achieved (per cent)

- 0% - 68%
- 70% - 204%
- 205% - 421%
- 422% - 529%
- No masons trained

Notes:

This map highlights the per cent of masons trained based on the targets defined in the paragraph below. The figure of masons trained through short training curriculum (7 days) was sourced from HRRP 4W-R21.

The target of short training (mason required per 15%) was sourced from a work-level target assessment survey conducted in Co-cha district by HRRP in Jan 2015.

Map File Name: TA_ShortTrainingCoverage_10x242
GLIDE Number: EO-2015-200043-NPL
Creation Date: 02 Dec 2016

Map Data Source: HRRP_4W_R21 (as of 25 Nov 2016)
Geo Data Source: MCFALD
Web Resource: <http://maptools.org/mw2/>



Disclaimer: The boundaries and names shown and the designations used in this map do not imply official endorsement or acceptance by the United Nations.



INDIA

CHINA

GORHA

RASUWA

DHADING

NUWAHAT

SINDHUPALCHOK

KATHMANDU

BHAKTAPUR

DOLAKHA

MAKWANPUR

LALITPUR

KAVREPALANCHOK

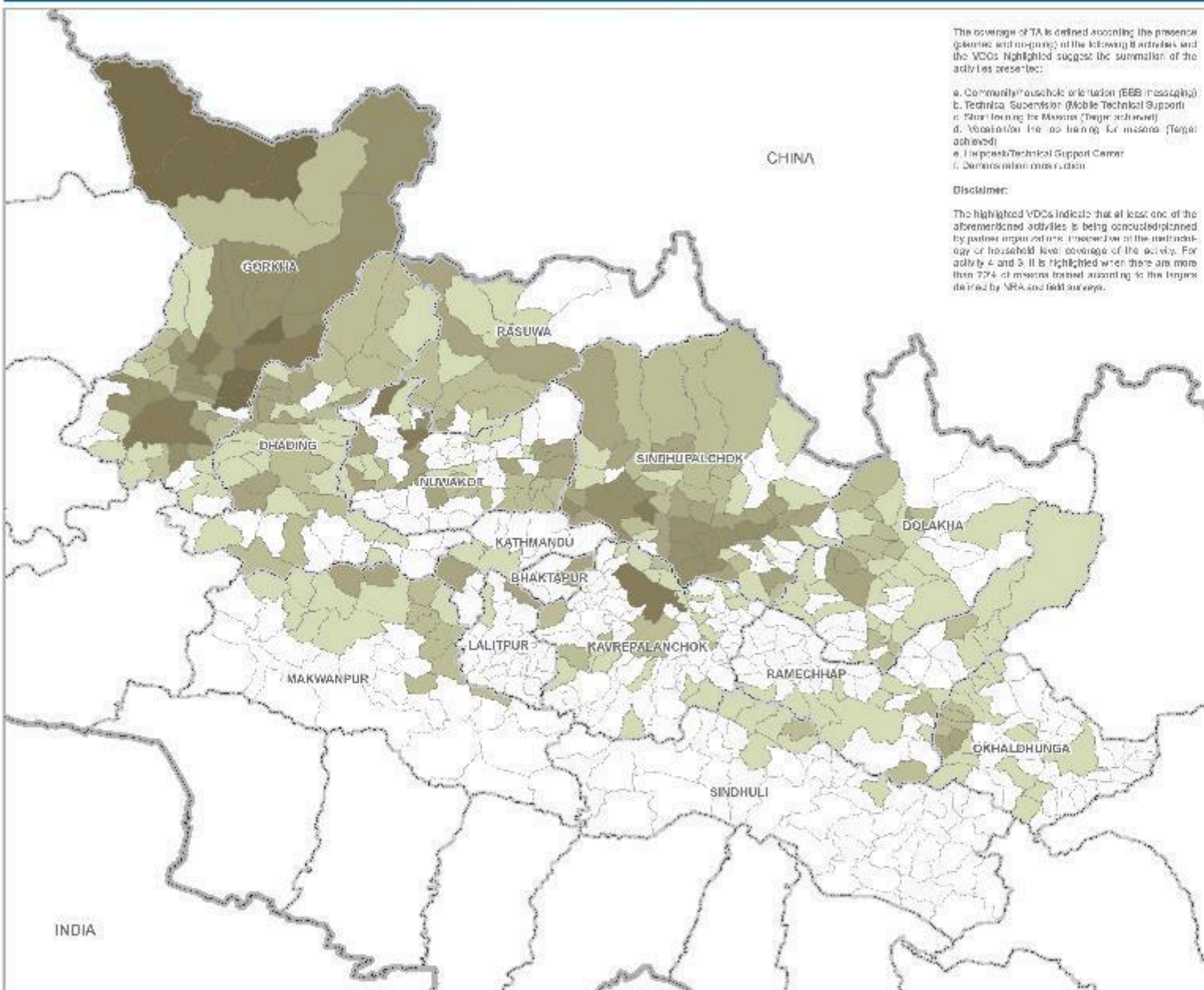
RAMECHHAP

OKHALDHUNGA

SINDHULI

Coverage

- Duplication, in some areas training at 889% of target
- In other areas no training at all has been delivered
- Quality assurance
- Other components of technical assistance important to ensure that masons have ongoing support during the reconstruction – gaps, partial coverage of VDCs, and duplication



Legend

- National Boundary
- District Boundary
- VDC/Village Boundary

Coverage of TA	VDCs/activities
1	139
2	52
3	55
4	30
5	10
6	9
Areas with no TA	205

Note:

This map highlights the coverage of Technical Assistance (TA) in Housing Sector by partner organizations in the 14 most affected districts.

Map Doc Name: TA_ActivityAnalysis_02244
 GIDP Number: EQ-2015-020945-VPL
 Creation Date: 02 Dec 2016

Map Data Source: HRRP_24_R21 (as of 25 Nov 2016)
 Geo Data Source: MapDL
 Web Resource: <http://mapdl.org/nepal/>



Disclaimer: The boundaries and names shown on this map do not imply official endorsement or acceptance by the United Nations.



Reconstruction Platform (HRRP)



VOCATIONAL TRAINING: PARTICIPANTS PERSPECTIVE

Information Collected

- What was the duration of the training course?
- Who conducted the training?
- How much do you charge households for your work (daily rate)?
- What are the challenges that you face in implementing the earthquake resistant construction elements covered in your training course?
- Are there any topics that your training didn't cover that you feel it should have? What are these?

Daily Rates

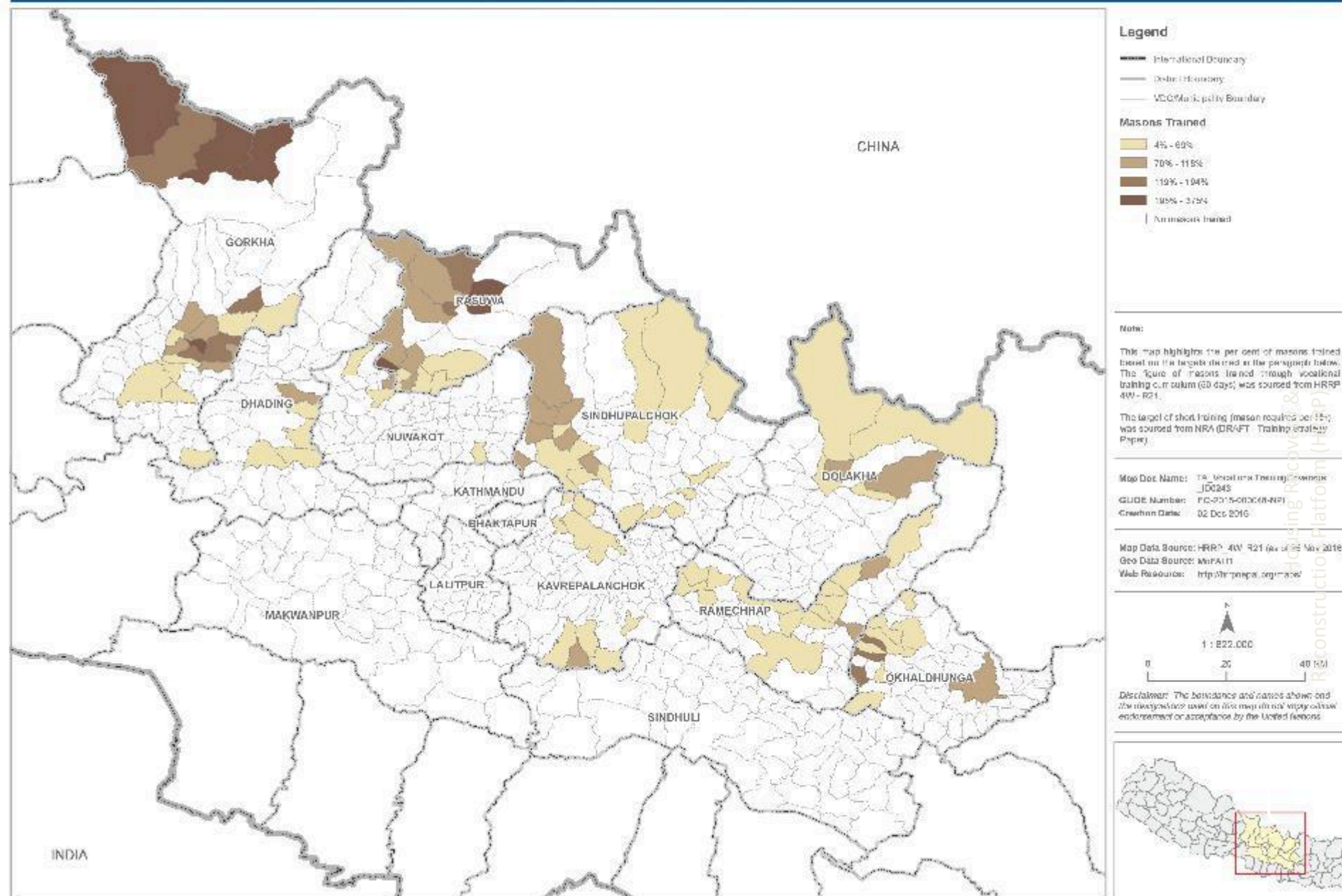
- **Dolakha:** Ranged from 600-700 NPRs / day, but this varies if food costs increase in the villages
- **Kavre:** Ranged from 800-1000 NPRs / day, and one reported a rate of 1000 NPRs excluding food. Increase of 200 NPRs reported.
- **Gorkha:** Ranged from 500-1000 NPRs / day and one reported that the labour price is by contract and is 200,000 NPRs for a 3 room house.
- **Nuwakot:** Ranged from 1000-1500 NPRs / day.
- **Okhaldhunga:** Reported 600 NPRs / day
- **Ramechhap:** Reported 600 NPRs / day with lunch or 700/800 NPRs / day without lunch.
- Rates are similar to existing masons rates

Challenges

- Shortage of stones and timber in the VDC
- Do not have advanced tools
- Lack of sufficient construction materials with the house owner and house owner making changes to the design provided by NRA
- People do not want to include banding because of the cost
- Training period was considered too short (both 30 and 50 day) and some trainees found it difficult to convince and gain trust of house owners and finding job is difficult
- Instead of a complete house construction, demo panels of each component were separately taught. So, while constructing a new house, there was some confusions and difficulties (30 days)
- Foundation detailing and bar bending is difficult with no guidance and supervision provided
- It is very difficult to teach other labourers about EQ resilient techniques
- Households don't want to follow the housing code and earthquake resistant techniques properly
- Don't have helmets and other tools for construction site

Recommendations

- Training should include more time on carpentry, steel reinforcement, RCC, roofing, reading and understanding drawings, and appropriate mortar and concrete mix
- It would be better if reference materials were provided on completion of training
- It would have been better if complete house was demonstrated step by step as is constructed in site (30 day training)
- Site layout was taught in a smaller scale. Had layout been taught at real scale, it would have been much easier
- Time duration was insufficient
- Refresher trainings would be good



Coverage

- Baliyo Ghar expanding to cover all gaps in Nuwakot, Dhading, and Dolakha
- JICA expanding to cover gaps in Gorkha and Sindhupalchowk
- Helvetas expanding working areas
- Indian Government, through UNDP, also looking to expand working areas in Nuwakot and Gorkha
- Need to **avoid duplication** and **focus on areas where there is no coverage**
- Since last Friday's meeting on TA, several organisations have contacted the HRRP re expanding their work on OJT



MASON'S PERSPECTIVE:
KAMAL BAHADUR SHRESTHA
WARD 01, BUNGKOT, GORKHA